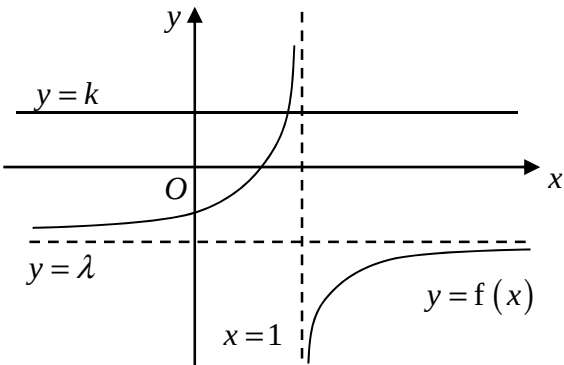
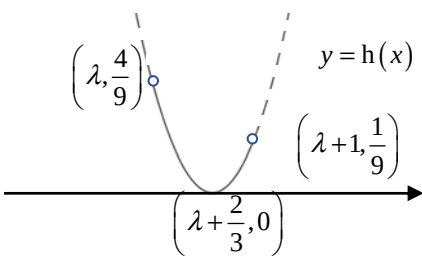
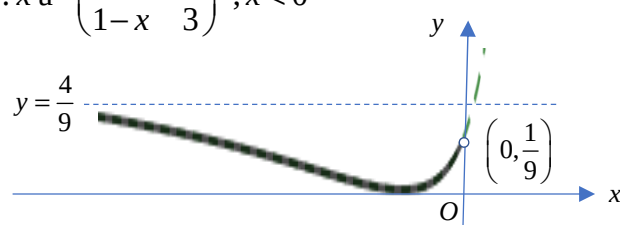


| Qn  | Solution                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1i  |  <p>Since <u>every</u> horizontal line <math>y = k</math>, <math>k \in \mathbb{R}_+</math>, cuts the graph of <math>y = f(x)</math> exactly once, <u>f is one-one</u>.<br/>Hence, <math>f^{-1}</math> exists.</p>                                                                                                                 |
| ii  | $\lambda + \frac{1}{1-x} = 2x$ $\lambda - \lambda x + 1 = 2x - 2x^2$ $2x^2 - (\lambda + 2)x + \lambda + 1 = 0$ $[-(\lambda + 2)]^2 - 4(2)(\lambda + 1) \geq 0$ $\lambda^2 + 4\lambda + 4 - 8\lambda - 8 \geq 0$ $\lambda^2 - 4\lambda - 4 \geq 0$ $\lambda \leq -0.82843 \text{ or } \lambda \geq 4.8284$ <p>Since <math>\lambda &lt; 0</math>, <math>\{\lambda \in \mathbb{R} : \lambda \leq -0.828\}</math>.</p> |
| iii | <p>Method A:<br/> <math>R_g = (\lambda, \lambda + 1)</math><br/>         Using <math>R_g</math> as the domain of <math>h</math>,</p>  <p>(Observe that the coordinates of the minimum point of <math>\left(x - \left(\lambda + \frac{2}{3}\right)\right)^2</math> are <math>\left(\lambda + \frac{2}{3}, 0\right)</math>.)</p>  |

| Qn | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | $R_{hg} = \left[ 0, \frac{4}{9} \right).$ <p>Method B:</p> $hg: x \mapsto \left( \frac{1}{1-x} - \frac{2}{3} \right)^2, x < 0$  $R_{hg} = \left[ 0, \frac{4}{9} \right).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2i | $\begin{aligned} & \cos\left(\frac{2n-1}{2}\right) - \cos\left(\frac{2n+1}{2}\right) \\ &= -2 \sin\left(\frac{1}{2}\left(\frac{2n-1}{2} + \frac{2n+1}{2}\right)\right) \\ & \quad \sin\left(\frac{1}{2}\left(\frac{2n-1}{2} - \frac{2n+1}{2}\right)\right) \\ &= -2 \sin\left(\frac{4n}{4}\right) \sin\left(-\frac{2}{4}\right) \\ &= -2 \sin(n) \left(-\sin\left(\frac{1}{2}\right)\right) \\ &= 2 \sin\left(\frac{1}{2}\right) \sin(n) \end{aligned}$ <p>Alternatively,</p> $\begin{aligned} 2 \sin\left(\frac{1}{2}\right) \sin(n) &= -(\cos P - \cos Q) \\ &= 2 \sin\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right) \end{aligned}$ <p>Let <math>\frac{P+Q}{2} = n</math> and <math>\frac{P-Q}{2} = \frac{1}{2}</math>.</p> $\begin{aligned} P+Q &= 2n \\ P-Q &= 1 \end{aligned}$ $2P = 2n+1 \Rightarrow P = \frac{2n+1}{2}$ $2Q = 2n-1 \Rightarrow Q = \frac{2n-1}{2}$ |

| Qn  | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | $2 \sin\left(\frac{1}{2}\right) \sin(n) = \cos Q - \cos P$ $= \cos\left(\frac{2n-1}{2}\right) - \cos\left(\frac{2n+1}{2}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| ii  | $\sum_{n=1}^N \sin(n)$ $= \sum_{n=1}^N \frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \left( 2 \sin\left(\frac{1}{2}\right) \sin(n) \right)$ $= \frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \sum_{n=1}^N \left( \cos\left(\frac{2n-1}{2}\right) - \cos\left(\frac{2n+1}{2}\right) \right)$ $= \frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \left( \begin{array}{l} \cos\left(\frac{1}{2}\right) - \cos\left(\frac{3}{2}\right) \\ + \cos\left(\frac{3}{2}\right) - \cos\left(\frac{5}{2}\right) \\ \quad \quad \quad \text{M} \\ + \cos\left(\frac{2N-3}{2}\right) - \cos\left(\frac{2N-1}{2}\right) \\ + \cos\left(\frac{2N-1}{2}\right) - \cos\left(\frac{2N+1}{2}\right) \end{array} \right)$ $= \frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \left( \cos\left(\frac{1}{2}\right) - \cos\left(\frac{2N+1}{2}\right) \right)$ $= \frac{1}{2} \cot\left(\frac{1}{2}\right) - \frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \cos\left(\frac{2N+1}{2}\right)$ |
| iii | <p>Since <math>\frac{2N+1}{2}</math> is not equal to odd multiples of <math>\frac{\pi}{2}</math> for all positive integers <math>N</math>,</p> $\frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \cos\left(\frac{2N+1}{2}\right) \neq 0.$ <p>Hence, the series will not equal <math>\frac{1}{2} \cot\left(\frac{1}{2}\right)</math> for all <math>N \in \mathbb{C}^+</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| iv  | $\sum_{n=10}^{25} \sin(n)$ $= \sum_{n=1}^{25} \sin(n) - \sum_{n=1}^9 \sin(n)$ $= \frac{1}{2} \cot\left(\frac{1}{2}\right) - \frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \cos\left(\frac{51}{2}\right) - \left( \frac{1}{2} \cot\left(\frac{1}{2}\right) - \frac{1}{2} \operatorname{cosec}\left(\frac{1}{2}\right) \cos\left(\frac{19}{2}\right) \right)$ $= -2.01$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Qn | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                             |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 3i |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                             |
| ii | $\int_3^5 f(x) dx$ $= \int_{-1}^1 2\sqrt{1+x^2} dx$ $= 2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{1+\tan^2 \theta} \sec^2 \theta d\theta$ $= 2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}  \sec \theta  \sec^2 \theta d\theta$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $x = \tan \theta$<br>$\frac{dx}{d\theta} = \sec^2 \theta$<br>When $x = -1, \theta = -\frac{\pi}{4}$<br>When $x = 1, \theta = \frac{\pi}{4}$ |
|    | <p>Since <math>\sec \theta &gt; 0</math> when <math>-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}</math>, <math> \sec \theta  = \sec \theta</math></p> $\int_3^5 f(x) dx = 2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^3 \theta d\theta$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                             |
|    | $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^3 \theta d\theta$ $= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 \theta \sec \theta d\theta$ $= \left[ \tan \theta \sec \theta \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec \theta \tan^2 \theta d\theta$ $= (\sqrt{2} + \sqrt{2}) - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec \theta (\sec^2 \theta - 1) d\theta$ $= 2\sqrt{2} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^3 \theta d\theta + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec \theta d\theta$ $2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^3 \theta d\theta = 2\sqrt{2} + \left[ \ln  \sec \theta + \tan \theta  \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$ $= 2\sqrt{2} + \ln(\sqrt{2} + 1) - \ln(\sqrt{2} - 1)$ $\int_3^5 f(x) dx = 2\sqrt{2} + \ln(\sqrt{2} + 1) - \ln(\sqrt{2} - 1)$ |                                                                                                                                             |

| Qn  | Solution                                                                                                                                                                                                                                                                                                                                                                                                                |                                |                 |                |   |     |            |   |               |                      |   |                         |                                |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------|----------------|---|-----|------------|---|---------------|----------------------|---|-------------------------|--------------------------------|
| 4i  | $\frac{20}{2}(2(2)+19t)=60$ $40+190t=60$ $t=\frac{2}{19}$ <p>If method A's claim is realistic, we will be able to recover more than 100% of the actual metal content.</p>                                                                                                                                                                                                                                               |                                |                 |                |   |     |            |   |               |                      |   |                         |                                |
| ii  | $\frac{10(1-r^{10})}{1-r}=60$ $1-r^{10}=6-6r$ $r^{10}-6r+5=0$ <p>If <math>r=1</math>, we will recover 100% after 10 reactions, instead of 60%.</p> <p>By GC, <math>r=0.879</math></p>                                                                                                                                                                                                                                   |                                |                 |                |   |     |            |   |               |                      |   |                         |                                |
| iii | <table><tr><th><math>n</math></th><th>Before reaction</th><th>After reaction</th></tr><tr><td>1</td><td>100</td><td><math>0.7(100)</math></td></tr><tr><td>2</td><td><math>0.7(100)+15</math></td><td><math>0.7^2(100)+15(0.7)</math></td></tr><tr><td>3</td><td><math>0.7^2(100)+15(0.7)+15</math></td><td><math>0.7^3(100)+15(0.7)^2+15(0.7)</math></td></tr></table> <p>Amount left after 3 reactions = 52.15 kg</p> | $n$                            | Before reaction | After reaction | 1 | 100 | $0.7(100)$ | 2 | $0.7(100)+15$ | $0.7^2(100)+15(0.7)$ | 3 | $0.7^2(100)+15(0.7)+15$ | $0.7^3(100)+15(0.7)^2+15(0.7)$ |
| $n$ | Before reaction                                                                                                                                                                                                                                                                                                                                                                                                         | After reaction                 |                 |                |   |     |            |   |               |                      |   |                         |                                |
| 1   | 100                                                                                                                                                                                                                                                                                                                                                                                                                     | $0.7(100)$                     |                 |                |   |     |            |   |               |                      |   |                         |                                |
| 2   | $0.7(100)+15$                                                                                                                                                                                                                                                                                                                                                                                                           | $0.7^2(100)+15(0.7)$           |                 |                |   |     |            |   |               |                      |   |                         |                                |
| 3   | $0.7^2(100)+15(0.7)+15$                                                                                                                                                                                                                                                                                                                                                                                                 | $0.7^3(100)+15(0.7)^2+15(0.7)$ |                 |                |   |     |            |   |               |                      |   |                         |                                |
| iv  | <p>Amount left after <math>n</math> reactions</p> $=(0.7)^n(100)+15(0.7+0.7^2+\cdots+0.7^{n-1})$ $=(0.7)^n(100)+\frac{15(0.7)(1-0.7^{n-1})}{1-0.7}$ $=(0.7)^n(100)+35(1-0.7^{n-1})$ $=70(0.7)^{n-1}+35-35(0.7)^{n-1}$ $=35(0.7)^{n-1}+35$ <p>When <math>n\rightarrow\infty</math>, <math>20(0.7)^{n-1}\rightarrow 0</math>.</p> <p>Amount left at the end of a reaction in the long run is 35 kg.</p>                   |                                |                 |                |   |     |            |   |               |                      |   |                         |                                |
| 5i  | $12!=479001600$                                                                                                                                                                                                                                                                                                                                                                                                         |                                |                 |                |   |     |            |   |               |                      |   |                         |                                |
| ii  | $2!\times 5!\times 6!=172800$                                                                                                                                                                                                                                                                                                                                                                                           |                                |                 |                |   |     |            |   |               |                      |   |                         |                                |
|     | $1\times 7!\times {}^8P_4=8467200$                                                                                                                                                                                                                                                                                                                                                                                      |                                |                 |                |   |     |            |   |               |                      |   |                         |                                |

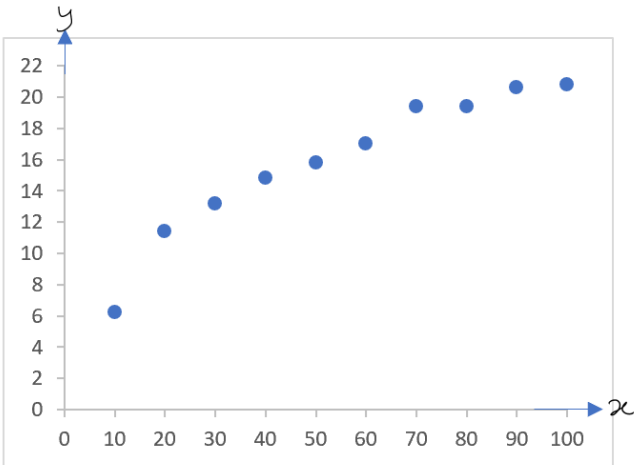
| Qn  | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | $1 - \frac{{}^6C_0 \times {}^6C_6 + {}^6C_1 \times {}^6C_5}{{}^{12}C_6} = \frac{887}{924} \text{ (or 0.960)}$                                                                                                                                                                                                                                                                                                                           |
| 6i  | $\bar{x} = 2$                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ii  | The probability that a frog carries the genetic trait is constant for all frogs <b>in a box</b> .<br>The presence of the genetic trait in a frog is independent among the frogs <b>in a box</b> .                                                                                                                                                                                                                                       |
| iii | $n = 6, p = \frac{1}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                |
|     | $X \sim B\left(6, \frac{1}{3}\right)$ $P(X = 2) = 0.32922$ <p>Let <math>Y</math> be the number of boxes that contain exactly 2 frogs with the genetic trait.</p> $Y \sim B(10, 0.32922)$ $P(Y > 3) = 1 - P(Y \leq 2) = 0.691$                                                                                                                                                                                                           |
| 7i  | <p>Let <math>y = x - 27</math></p> <p>Unbiased estimate of population mean <math>= \bar{x}</math></p> $= \bar{y} + 27$ $= \frac{\sum y}{n} + 27$ $= \frac{12}{50} + 27$ $= 27.24$ <p>Unbiased estimate of population variance <math>= s^2</math></p> $= \frac{1}{n-1} \left( \sum y^2 - \frac{(\sum y)^2}{n} \right)$ $= \frac{1}{49} \left( 36.4 - \frac{12^2}{50} \right)$ $= \frac{33.52}{49}$ $= 0.68408$ $= 0.684 \text{ (3 s.f)}$ |
| ii  | <p>Let <math>X</math> cm be the length of a steel rod.</p> <p>Let <math>\mu</math> cm be the population mean length of steel rods.</p>                                                                                                                                                                                                                                                                                                  |

| Qn  | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p> <math>H_0 : \mu = 27</math><br/> <math>H_1 : \mu \neq 27</math><br/> Level of significance: 2.5%<br/> Test Statistic: since <math>n = 50</math> is sufficiently large, by Central Limit Theorem, <math>\bar{X}</math> is approximately normal.<br/> When <math>H_0</math> is true,<br/> <math display="block">Z = \frac{\bar{X} - 27}{\frac{s}{\sqrt{50}}} : N(0,1) \text{ approximately}</math> Computation: <math>\bar{x} = 27.24</math>, <math>s = \sqrt{0.68408} = 0.82709</math>,<br/> <math>p\text{-value} = 0.040185 = 0.0402</math> (3 s.f)<br/> Conclusion : Since <math>p\text{-value} = 0.0402 &gt; 0.025</math>, <math>H_0</math> is not rejected at 2.5% level of significance. Hence there is insufficient evidence that the population mean length of steel rods produced differs from 27.0 cm.<br/><br/> Since <math>n = 50</math> is large enough, by Central Limit Theorem, sample mean lengths of steel rods is approximately normal. Hence, there is no need for the population distribution of lengths of steel rods to be known. </p> |
| iii | <p>There is a probability of 0.025 that the test will conclude that the population mean length of steel rods produced differs from 27.0 cm when it is actually 27.0 cm.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|     | <p> <math>H_0 : \mu = l</math><br/> <math>H_1 : \mu &gt; l</math><br/> Level of significance: 5%<br/> Test Statistic: since <math>n = 50</math> is sufficiently large, by Central Limit Theorem, <math>\bar{X}</math> is approximately normal.<br/> When <math>H_0</math> is true,<br/> <math display="block">Z = \frac{\bar{X} - l}{\frac{3}{\sqrt{50}}} : N(0,1) \text{ approximately .}</math> Computation:<br/> To reject <math>H_0</math>, <math>z \geq 1.6449</math><br/> <math display="block">\frac{23.8 - l}{\frac{3}{\sqrt{50}}} \geq 1.6449</math> <math display="block">l \leq 23.102</math><br/><br/> <math>\{l \in \mathbb{R}^+ : l \leq 23.1\}</math> </p>                                                                                                                                                                                                                                                                                                                                                                                       |

| Qn       | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                        |                               |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------|-------|---------|----------|--------------------------------|--------------------------------------------------------|-------------------------------|--|---------------------|--------------------|--------------------|-----|-------|------|---|----------|--------------------|-------------------|-------------------|
| 8        | Let the probability of obtaining a '6' be $k$ .<br>$5\left(\frac{1}{3}k\right)+k=1$ $\frac{8}{3}k=1\Rightarrow k=\frac{3}{8}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                        |                               |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| i        | $P(\text{obtaining a '6'})=\frac{1}{4}\times\frac{3}{8}+\frac{3}{4}\times\frac{1}{6}$ $=0.21875\left(\text{or }\frac{7}{32}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                        |                               |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| ii       | $P(\text{obtaining } n \text{ '6's})=\frac{1}{4}\times\left(\frac{3}{8}\right)^n+\frac{3}{4}\times\left(\frac{1}{6}\right)^n$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                        |                               |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| b        | $\frac{\frac{1}{4}\times\left(\frac{3}{8}\right)^n}{\frac{1}{4}\times\left(\frac{3}{8}\right)^n+\frac{3}{4}\times\left(\frac{1}{6}\right)^n}>0.95$ <table border="1"><thead><tr><th><math>n</math></th><th>Probability</th></tr></thead><tbody><tr><td>4</td><td>0.89521</td></tr><tr><td>5</td><td>0.95055</td></tr></tbody></table> <p>Least <math>n</math> is 5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $n$                                                    | Probability                   | 4     | 0.89521 | 5        | 0.95055                        |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| $n$      | Probability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                        |                               |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| 4        | 0.89521                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                        |                               |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| 5        | 0.95055                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                        |                               |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| iii      | Let $\$X$ be the winnings of the player. <table border="1"><thead><tr><th><math>x</math></th><th><math>-2k</math></th><th><math>2-k</math></th><th>4</th></tr></thead><tbody><tr><td><math>P(X=x)</math></td><td><math>\left(\frac{25}{32}\right)^2</math></td><td><math>2\left(\frac{7}{32}\right)\left(\frac{25}{32}\right)</math></td><td><math>\left(\frac{7}{32}\right)^2</math></td></tr><tr><td></td><td><math>=\frac{625}{1024}</math></td><td><math>=\frac{175}{512}</math></td><td><math>=\frac{49}{1024}</math></td></tr></tbody></table> $E(X)=-2k\left(\frac{625}{1024}\right)+(2-k)\left(\frac{175}{512}\right)+4\left(\frac{49}{1024}\right)$ $0=-\frac{625k}{512}+\frac{175}{256}-\frac{175k}{512}+\frac{49}{256}$ $\frac{25}{16}k=\frac{7}{8}$ $k=0.56$ <table border="1"><thead><tr><th><math>x</math></th><th>-1.12</th><th>1.44</th><th>4</th></tr></thead><tbody><tr><td><math>P(X=x)</math></td><td><math>\frac{625}{1024}</math></td><td><math>\frac{175}{512}</math></td><td><math>\frac{49}{1024}</math></td></tr></tbody></table> $\text{Var}(X)=2.24$ <p>The variance of the player's winnings is \$<sup>2</sup>2.24.</p> | $x$                                                    | $-2k$                         | $2-k$ | 4       | $P(X=x)$ | $\left(\frac{25}{32}\right)^2$ | $2\left(\frac{7}{32}\right)\left(\frac{25}{32}\right)$ | $\left(\frac{7}{32}\right)^2$ |  | $=\frac{625}{1024}$ | $=\frac{175}{512}$ | $=\frac{49}{1024}$ | $x$ | -1.12 | 1.44 | 4 | $P(X=x)$ | $\frac{625}{1024}$ | $\frac{175}{512}$ | $\frac{49}{1024}$ |
| $x$      | $-2k$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $2-k$                                                  | 4                             |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| $P(X=x)$ | $\left(\frac{25}{32}\right)^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $2\left(\frac{7}{32}\right)\left(\frac{25}{32}\right)$ | $\left(\frac{7}{32}\right)^2$ |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
|          | $=\frac{625}{1024}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $=\frac{175}{512}$                                     | $=\frac{49}{1024}$            |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| $x$      | -1.12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1.44                                                   | 4                             |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |
| $P(X=x)$ | $\frac{625}{1024}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\frac{175}{512}$                                      | $\frac{49}{1024}$             |       |         |          |                                |                                                        |                               |  |                     |                    |                    |     |       |      |   |          |                    |                   |                   |



| Qn                                                                                                                  | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                     |                                                                                                                   |     |            |                 |                |     |         |     |            |                 |                |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----|------------|-----------------|----------------|-----|---------|-----|------------|-----------------|----------------|
|                                                                                                                     | <p>Alternatively,<br/>Let \$A be the winnings from randomly choosing one dice from the bag and rolled once.</p> <p>Let \$B be the winnings of the player.<br/><math>B = A_1 + A_2</math></p> $E(B) = E(A_1 + A_2)$ $= 2E(A)$ <p>Given that the expected winnings is \$0,<br/><math>E(B) = 0 \Rightarrow 2E(A) = 0 \Rightarrow E(A) = 0</math></p> <table><tr><td><math>a</math></td><td><math>-k</math></td><td><math>2</math></td></tr><tr><td><math>P(A = a)</math></td><td><math>\frac{25}{32}</math></td><td><math>\frac{7}{32}</math></td></tr></table> $\frac{25}{32}(-k) + \frac{7}{32}(2) = 0$ $\Rightarrow k = 0.56 \text{ (shown)}$ $\text{Var}(B) = \text{Var}(A_1 + A_2)$ $= 2\text{Var}(A)$ <table><tr><td><math>a</math></td><td><math>-0.56</math></td><td><math>2</math></td></tr><tr><td><math>P(A = a)</math></td><td><math>\frac{25}{32}</math></td><td><math>\frac{7}{32}</math></td></tr></table> <p>By GC, <math>\text{Var}(A) = 1.12</math><br/><math>\text{Var}(B) = 2.24</math><br/>The variance of the player's winnings is \$2.24.</p> | $a$                                                                                                                 | $-k$                                                                                                              | $2$ | $P(A = a)$ | $\frac{25}{32}$ | $\frac{7}{32}$ | $a$ | $-0.56$ | $2$ | $P(A = a)$ | $\frac{25}{32}$ | $\frac{7}{32}$ |
| $a$                                                                                                                 | $-k$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $2$                                                                                                                 |                                                                                                                   |     |            |                 |                |     |         |     |            |                 |                |
| $P(A = a)$                                                                                                          | $\frac{25}{32}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\frac{7}{32}$                                                                                                      |                                                                                                                   |     |            |                 |                |     |         |     |            |                 |                |
| $a$                                                                                                                 | $-0.56$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $2$                                                                                                                 |                                                                                                                   |     |            |                 |                |     |         |     |            |                 |                |
| $P(A = a)$                                                                                                          | $\frac{25}{32}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\frac{7}{32}$                                                                                                      |                                                                                                                   |     |            |                 |                |     |         |     |            |                 |                |
| 9i                                                                                                                  | <p>Let <math>X</math> g be the mass of a randomly chosen Cameo Apple.<br/><math>X \sim N(\mu, \sigma^2)</math></p> <table><tr><td><math>P(X &gt; 125) = 0.081</math><br/><math>P\left(Z &gt; \frac{125 - \mu}{\sigma}\right) = 0.081</math><br/><math>\frac{125 - \mu}{\sigma} = 1.3984</math></td><td><math>P(X &lt; 90) = 0.147</math><br/><math>P\left(Z &gt; \frac{90 - \mu}{\sigma}\right) = 0.147</math><br/><math>\frac{90 - \mu}{\sigma} = -1.0494</math></td></tr></table> <p>By GC,<br/><math>\mu = 105.00, \quad \sigma = 14.298</math><br/><math>\approx 105 \quad \approx 14.3</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $P(X > 125) = 0.081$<br>$P\left(Z > \frac{125 - \mu}{\sigma}\right) = 0.081$<br>$\frac{125 - \mu}{\sigma} = 1.3984$ | $P(X < 90) = 0.147$<br>$P\left(Z > \frac{90 - \mu}{\sigma}\right) = 0.147$<br>$\frac{90 - \mu}{\sigma} = -1.0494$ |     |            |                 |                |     |         |     |            |                 |                |
| $P(X > 125) = 0.081$<br>$P\left(Z > \frac{125 - \mu}{\sigma}\right) = 0.081$<br>$\frac{125 - \mu}{\sigma} = 1.3984$ | $P(X < 90) = 0.147$<br>$P\left(Z > \frac{90 - \mu}{\sigma}\right) = 0.147$<br>$\frac{90 - \mu}{\sigma} = -1.0494$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                     |                                                                                                                   |     |            |                 |                |     |         |     |            |                 |                |
| ii                                                                                                                  | Let $C$ g be the mass of a randomly bag of Cameo apples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                     |                                                                                                                   |     |            |                 |                |     |         |     |            |                 |                |

| Qn      | Solution                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | $C = X_1 + X_2 + K + X_8$ $C \sim N(840, 1635.92)$ $P(C < 845) = 0.54919 \approx 0.549 \text{ (3sf)}$                                                                                                                                                                                                                                                                                             |
| iii     | <p>Let <math>J</math> g be the mass of a randomly chosen bag of Jazz apples.</p> $Y = \frac{7.30}{1000}(J_1 + J_2 + J_3) - \frac{8.50}{1000}(2C)$ $E(Y) = \frac{7.30}{1000}(3)(700) - \frac{8.50}{1000}(2)(840)$ $= 1.05$ $\text{Var}(Y)$ $= \left(\frac{7.30}{1000}\right)^2 (3)(35)^2 + \left[\frac{8.50}{1000}(2)\right]^2 (1635.92)$ $= 0.66862$ $Y \sim N(1.05, 0.66862)$ $P(Y > 1) = 0.524$ |
| 10<br>i |                                                                                                                                                                                                                                                                                                                |
| ii a    | $r = 0.9541$ (4 d.p)                                                                                                                                                                                                                                                                                                                                                                              |
| b       | $r = 0.9942$ (4 d.p)                                                                                                                                                                                                                                                                                                                                                                              |
| iii     | <p>From the scatter diagram, as <math>x</math> increases, <math>y</math> increases by decreasing amounts. In addition, the product moment correlation coefficient between <math>\ln x</math> and <math>y</math>, 0.9942, is closer to 1 as compared to that between <math>x</math> and <math>y</math>, 0.9541. Hence <math>y = c + d \ln x</math> is the better model.</p>                        |
| iv      | <p>Since <math>x</math> is the independent variable, neither the regression line of <math>x</math> on <math>y</math> nor the regression line of <math>\ln x</math> on <math>y</math> should be used to estimate the value of <math>x</math> when <math>y = 17.2</math>.</p>                                                                                                                       |
| v       | Equation of regression line of $y$ on $\ln x$ is                                                                                                                                                                                                                                                                                                                                                  |

| Qn | Solution                                                                                                                                     |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|
|    | $y = 6.3074 \ln x - 8.1904$ $y = 6.31 \ln x - 8.19$ <p>When <math>y = 17.2</math>,</p> $17.2 = -8.1904 + 6.3074 \ln x$ $x = 56.008$ $= 56.0$ |